222 g of $Mg_2P_2O_7$ contains 62 g of P

0.66 g contains $\left(\frac{62}{222} \times 0.66\right)$ g of P

Percentage of P $\frac{62}{222} \times \frac{0.66}{0.24} \times 100 = 76.80\ \%$

Evaluate Yourself

9) 0.33 g of an organic compound containing phosphorous gave 0.397 g of $Mg_2P_2O_7$ by the analysis. Calculate the percentage of P in the compound (Ans: 23.21) (MFW of $Mg_2P_2O_7$ is 222 P = 31)

Estimation of nitrogen: There are two methods for the estimation of nitrogen in an organic compound. They are 1. Dumas method 2. Kjeldahls method

1. Dumas method:

This method is based upon the fact that nitrogenous compound when heated with cupric oxide in an atmosphere of CO_2 yields free nitrogen. Thus

$$C_x H_y N_z + (2x + y/2)\ CuO \longrightarrow x\ CO_2 + y/2\ H_2O + z/2\ N_2 + (2x + y/2)\ Cu$$

Traces of oxide of nitrogen, which may be formed in some cases, are reduced to elemental nitrogen by passing over heated copper spiral.

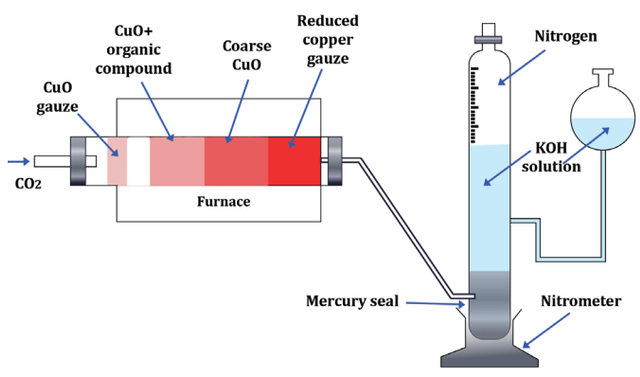


Fig 11.4 Dumas Method

The apparatus used in Dumas method consists of CO_2 generator, combustion tube, Schiffs nitrometer. (See Fig. 11.4)

CO_2 generator:

CO_2 needed in this process is prepared by heating magnetite or sodium bicarbonate contained in a hard glass tube or by the action of dil. HCl on marble in a kipps apparatus. The gas is passed through the combustion tube after being dried by bubbling through Conc. H_2SO_4.

Combustion Tube: The combustion tube is heated in a furnace is charged with a) A roll of oxidized copper gauze to prevent the back diffusion of the products of combustion and to heat the organic substance mixed with CuO by radiation b) a weighed amount of the organic substance mixed with excess of CuO, C) a layer of course CuO packed in about 2/3 of the entire length of the tube and kept in position by loose asbestos plug on either side; this oxidizes the organic vapors passing through it, and d) a reduced copper spiral which reduces any oxides of nitrogen formed during combustion to nitrogen.

Schiff's nitro meter: The nitrogen gas obtained by the decomposition of the substance in the combustion tube is mixed with considerable excess of CO_2 It is estimated by passing nitrometer when CO_2 is absorbed by KOH and the nitrogen gets collected in the upper part of graduated tube.

Procedure: To start with the tap of nitro meter is left open. CO_2 is passed through the combustion tube to expel the air in it. When the gas bubbles rising through, the

Unit 11 revised - jagan.indd 145 22-01-2021 12:42:30